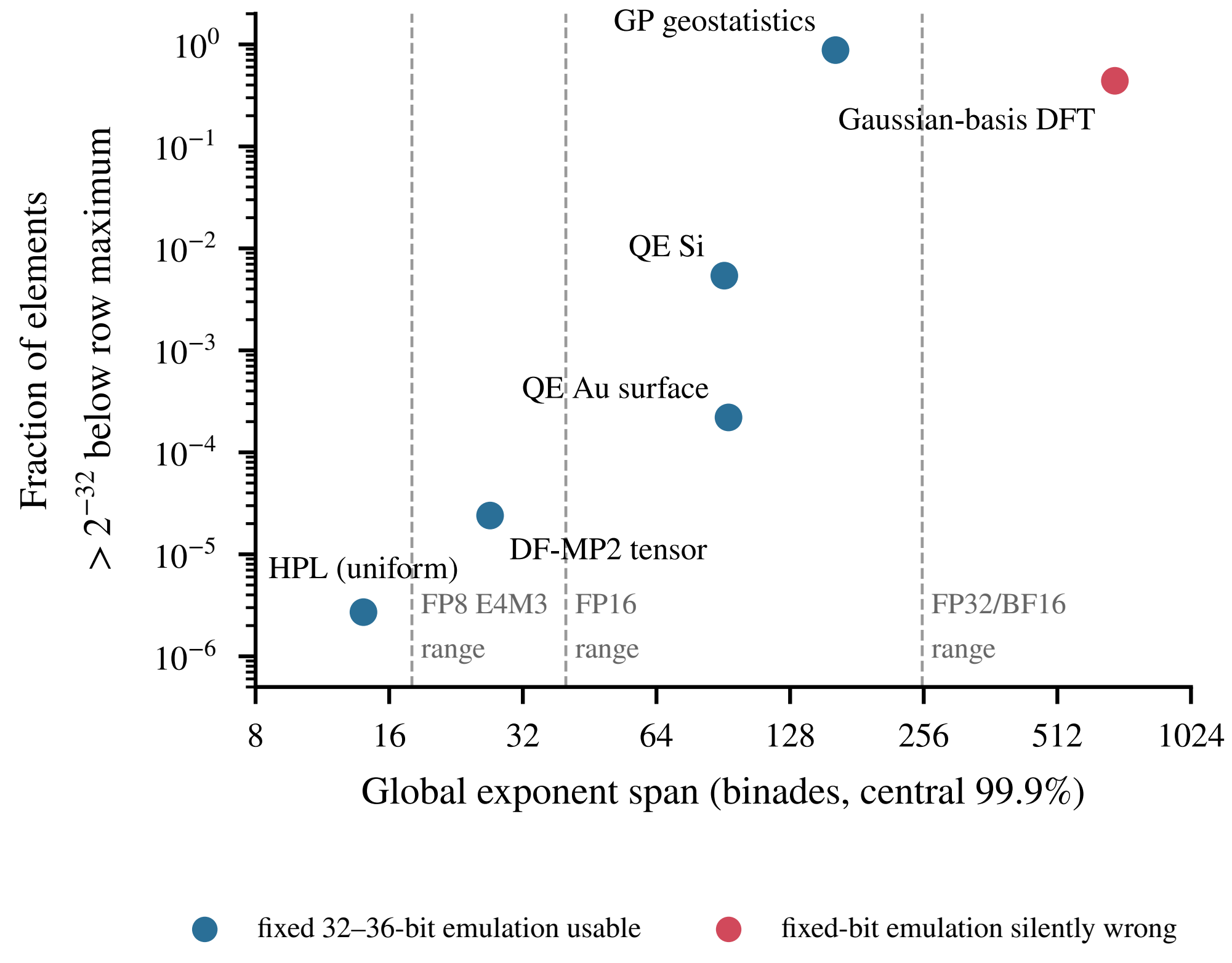
GP geostatistics
Gaussian-basis DFT
QE Si
QE Au surface
DF-MP2 tensor
HPL (uniform)
FP8 E4M3 range
FP16 range
FP32/BF16 range
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
Fraction of elements $> 2^{-32}$ below row maximum
8
16
32
64
128
256
512
1024
Global exponent span (binades, central 99.9%)
fixed 32–36-bit emulation usable
fixed-bit emulation silently wrong